From tinySA

Main: USB Interface

After connecting the tinySA to a PC using the a USB cable a serial interface will become available on the PC. Drivers for the serial interface over USB will be automatically installed on Windows and are already builtin in most Linux kernels The serial interface supports a large set of commands. You can always use the **help** command to get a list of the available commands

There is limited error checking against incorrect parameters of incorrect mode

- Frequencies can be specified using an integer optionally postfixed with a the letter 'k' for kilo 'M' for Mega or 'G' for Giga. E.g. 0.1M (100kHz), 500k (0.5MHz) or 12000000 (12MHz)
- Levels are specified in dB(m) and can be specified using a floating point notation. E.g. 10 or 2.5
- Time is specified in seconds optionally postfixed with the letters 'm' for mili or 'u' for micro. E.g. 1 (1 second), 2.5 (2.5 seconds), 120m (120 milliseconds)

Commands:

attenuate
: sets the internal attenuation to automatic or a specific value
: usage: attenuate [auto|0-31]

abort
: enables the previous command abortion of aborts the previous command
: usage: abort [off| on]
: When used without parameters the previous command still running will be aborted. Abort must be enabled before usage using the "abort on" command

bulk
: send by tinySA when in auto refresh mode
: format: "bulk\r\n{X}{Y}{Width}{Height}{Pixeldata}\r\n"
: where all numbers are binary coded 2 bytes little endian. The Pixeldata is encoded as 2 bytes per pixel

calc
: sets or cancels one of the measurement modes
: usage: calc off|minh|maxh|maxd|aver4|aver16|quasip
: the commands are the same as those listed in the MEASURE menu

caloutput
: disables or sets the caloutput to a specified frequency in MHz
: usage: caloutput off|30|15|10|4|3|2|1

capture
: requests a screen dump to be send in binary format of 320x240 pixels of each 2 bytes

clearconfig
: resets the configuration data to factory defaults
: usage: clearconfig 1234

color
: sets or dumps the colors used
: usage: color [{id} {rgb24}]

correction
: sets or dumps the frequency level correction table
usage: correction {table_name} [(0..9|0..19) {frequency} {level}]

dac
: sets or dumps the dac value
usage: dac [0..4095]

data
: dumps the trace data
usage: data 0..2
0=temp value, 1=stored trace, 2=measurement

deviceid
: sets of dumps a user settable number that can be use to identify a specific tinySA
usage: deviceid [{number}]

fill
: send by tinySA when in auto refresh mode
format: "fill\r\n{X}{Y}{Width}{Height}{Color}\r\n"
where all numbers are binary coded 2 bytes little endian.

freq
: pauses the sweep and sets the measurement frequency
usage: freq {frequency}

frequencies
: dumps the frequencies used by the last sweep
usage: frequencies

if
: sets the IF to automatic or a specific value
usage: if (0 | 433M..435M)
where 0 means automatic

info
: displays various SW and HW information

help
: dumps a list of the available commands
usage: help

hop (Ultra only)
: measures the input level at each of the requested frequencies
usage: hop {start(Hz)} {stop(Hz)} {step(Hz) | points} [outmask]
If the 3rd parameter is below 450 it is assumed to be points, otherwise as step frequency
Outmask selects if the frequency (1) or level (2) is output.

level
: sets the output level
usage: level -76..13
Not all values in the range are available

levelchange
: sets the output level delta for low output mode level sweep
usage: levelchange -70..+70

leveloffset
: sets or dumps the level calibration data
usage: leveloffset low|high|switch [output] {error}
For the output corrections first ensure correct output levels at maximum output level. For the low output set the output to -50dBm and measure and correct the level with "leveloffset switch error" where
For all output leveloffset commands measure the level with the leveloffset to zero and calculate error = measured level - specified level

load
: loads a previously stored preset
usage: load 0..4
where 0 is the startup preset

marker
: sets or dumps marker info
usage: marker {id} on|off|peak|{freq}|{index}
where id=1..4 index=0..num_points-1
Merker levels will use the selected unit
Marker peak will activate the marker (if not done already), position the marker on the strongest signal and display the marker info
The frequency must be within the selected sweep range

menu
: The menu command can be used to activate any menu item.
usage: menu {#} [{#} [{#} [{#}]]]
where # is the menu entry number starting with 1 at the top.
Example: menu 6 2 will toggle the waterfall option

mode
: sets the mode of the tinySA
usage: mode low|high input|output

modulation
: sets the modulation in output mode
usage: modulation off|AM_1kHz|AM_10Hz|NFM|WFM|extern

ext_gain
: sets the external attenuation/amplification
usage: ext_gain -100..100
Works in both input and output mode

output
: sets the output on or off
usage: output on|off

pause
: pauses the sweeping in either input or output mode
usage: pause

rbw
: sets the rbw to either automatic or a specific value
usage: rbw auto|3..600
the number specifies the target rbw in kHz

recall

same as load

refresh
: enables/disables the auto refresh mode
Usage refresh on|off

release
: signals a removal of the touch
usage: release

remark
: does nothing
usage: remark use any text

repeat
: sets the number of measurements that should be taken at every frequency
usage: repeat 1..1000
increasing the repeat reduces the noise per frequency, repeat 1 is the normal scanning mode.

reset
: resets the tinySA
usage: reset

restart
: restarts the the tinySA after the specified number of seconds
usage: restart {seconds}
where 0 seconds stops the restarting.

resume
: resumes the sweeping in either input or output mode
usage: resume

save
: saves the current setting to a preset
usage: save 0..4
where 0 is the startup preset

saveconfig
: saves the device configuration data
usage: saveconfig

scan
: performs a scan and optionally outputs the measured data
usage: scan {start(Hz)} {stop(Hz)} [points] [outmask]
where the outmask is a binary OR of 1=frequencies, 2=measured data, 4=stored data and points is maximum 290

scanraw
: performs a scan of unlimited amount of points and send the data in binary form
usage: scanraw {start(Hz)} {stop(Hz)} [points] [option]
The measured data is the level in dBm and is send as '{' ('x' MSB LSB)*points '}'. To get the dBm level from the 16 bit data, divide by 32 and subtract 128 for the tinySA and 174 for the tinySA Ultra.
The option, when present, can be either 0,1,2 or 3 being the sum of 1=unbuffered and 2=continuous

selftest

performs one or all selftests
usage: selftest 0 0..9

spur
enables or disables spur reduction
usage: spur on|off

sweep
set sweep boundaries or execute a sweep
usage: sweep [(start|stop|center|span|cw {frequency}) | ({start(Hz)} {stop(Hz)} [0..290])]
sweep without arguments lists the current sweep settings, the frequencies specified should be within the permissible range. The sweep commands apply both to input and output modes

sweeptime
sets the sweeptime
usage: sweep {time(Seconds)}the time specified may end in a letter where m=mili and u=micro

text
specifies the text entry for the active keypad
usage: text keypadtext
where keypadtext is the text used
Example: text 12M
Currently does not work for entering file names

threads
lists information of the threads in the tinySA

touch
sends the coordinates of a touchusage: touch {X coordinate} {Y coordinate}
The upper left corner of the screen is "0 0"

touchcal
starts the touch calibration

touchtest
starts the touch test

trace
displays all or one trace information or sets trace related information
usage: trace [{0..2} | dBm|dBmV|dBuV|V|W |store|clear|subtract | (scale|reflevel) auto|{level}

trigger
sets the trigger type or level
usage: trigger auto|normal|single|{level(dBm)}
the trigger level is always set in dBm

vbat
displays the battery voltage

vbat_offset
displays or sets the battery offset value
usage: vbat_offset [{0..4095}]

version
displays the version text

wait
: wait for a single sweep to finish and pauses sweep or waits for specified number of seconds
usage: wait [{seconds}]

waitscan
: wait for the specified number of scans, or the current scan if not specified, to conclude and leaves the device in pause mode. All command execution will be clocked during waiting for the scans to complete
usage: waitscan [{scans}]
